The missing pressure traction in Scott–Vogelius–Nitsche methods on polygonal approximations of a curved wall

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Abstract

We study smooth two-dimensional Stokes flow with a curved no-slip wall approximated by an inscribed polygon, using Scott–Vogelius elements of degree $k \geq 4$ and Nitsche's method. For the printed Gjerde–Scott formulation, the pressure traction is absent. Under the approximation assumptions of Theorem 1(c), with fixed effective penalty, bounded mesh ratio and nonconstant wall pressure, the leading normal leak is $(h_\Omega/\mu)(p-\bar{p})$, with asymptotic constant 1. The energy and H^1 errors have orders $(h_\Omega/\mu)^{1/2}$ and h_Ω/μ, respectively, with matching lower bounds. A mean-free pressure-traction correction has the same velocity as a zero-net-flux formulation of Frachon, Nilsson and Zahedi and attains $O(h_\Gamma^{3/2} + h_\Omega^k)$ under the stated assumptions and a sufficiently large, bounded effective penalty. Under condition (S) and nonzero wall shear, the exponent $3/2$ is sharp for every $k \geq 4$. A penalty proportional to $h_\Omega/\min_e |e|$ is sufficient and, under a triangle-shape condition, necessary. Quartic computations support the predicted rates, leak constant and penalty thresholds.

Keywords: Stokes equations, Scott–Vogelius elements, Nitsche's method, curved boundaries, penalty parameter

1. Introduction

In 1985 Scott and Vogelius showed that for polynomial degree $k \geq 4$, on meshes without singular vertices, the divergence maps continuous piecewise-P_k velocities onto discontinuous piecewise-P_{k-1} pressures [1, 2]. The velocities are then divergence-free pointwise, with exact mass conservation and pressure-robust errors on fitted meshes [3]. At a curved wall the same property causes trouble. If the wall is replaced by an inscribed polygon and no-slip is imposed strongly, the divergence constraint can force a zero velocity gradient at wall vertices with two incident triangles. A positive fraction of such locking vertices, in a region with nonzero wall shear, can lead to an H^1 error of order $h_\Gamma^{1/2}$ [4, 5]. Under uniform shape and angle control, including (M2), and compatible outer data, strong imposition can also attain $h_\Gamma^{3/2} + h_\Omega^k$ [6]. Weak imposition therefore remains relevant on meshes exhibiting locking or approaching degenerate wall stars. Gjerde and Scott [5] therefore keep the polygon but impose the wall condition weakly, by Nitsche's method [7], with penalty μ/h_Ω on a mesh of size h_Ω. For a zero-pressure benchmark Scott observed H^1 errors of order $h^{3/2}$, the rate expected from an uncorrected polygon (corrections beyond it for scalar problems go back to [8]; see [9]). For Scott's constant-pressure benchmark $u = (1-r^{-2})(-y, x)$, $p = 0$, Cavalcante proved $\|u - u_h\|_{H^1(\Omega_h)} \leq C(h_\Omega^k + h_\Gamma^{3/2})$ for every fixed $k \geq 4$, under the stated geometric and uniform mesh assumptions, flux-compatible outer data, and a sufficiently large, bounded effective-penalty window $\gamma_0 \leq \mu h_\Gamma/h_\Omega \leq \gamma_1$ [10, Theorem 5.2]. The constant is independent of the two mesh scales. Scott offered a prize for a proof of the error $h_\Gamma^{3/2} + h_\Omega^k$, h_Γ the longest polygon edge, for general Stokes data: "If this does not hold, why not?" [4].

For the method as printed it does not hold, because a term is missing. The consistent Nitsche form for Stokes flow uses the full traction, so it contains $\langle p, v \cdot n \rangle$ [11, 12]; the form printed in [5, (27)–(28)] omits it. Without a traction

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only the penalty balances the wall pressure, and the discrete flow leaks through the polygon with normal velocity $u_h \cdot n \approx (h_\Omega/\mu)(p - \bar{p})$, where $\bar{p}$ is the mean of p on the wall (the mean cannot leak, as the net flux is zero). This is the consistency error of penalty methods [13], acting on the pressure alone: for fixed μ the H^1 rate is 1. Restoring the full traction fails for divergence-free spaces: the constant pressure mode then drops out of the equations and is not determined. Subtracting the wall mean of the discrete pressure repairs this. The velocity of the resulting method is that of the zero-net-flux formulation of Frachon, Nilsson and Zahedi [14, §5.1], transcribed to this setting.

Related work and novelty. For the constant-pressure benchmark, there is no pressure-driven leak. Cavalcante's estimate also holds for every fixed $k \geq 2$ on admissible barycentric Clough–Tocher refinements of uniformly shape-regular macrotriangulations, with the same geometric assumptions, flux-compatible outer data and penalty window, using the degree-k spaces and trace interpolant [10, Corollary 5.3]. Exactly divergence-free methods of the optimal order h^k exist on curved domains, at a higher cost: isoparametric Piola-mapped Scott–Vogelius elements fit the wall and are pressure-robust [15, Theorem 6.1], also in 3D [16], and Liu, Neilan and Otus [17] work inside the curved domain with a boundary multiplier, a zero-flux constraint and a Taylor correction. The placement of the pressure traction in divergence-free cut methods is discussed in [18, 14]; our stability theory is that of Guzmán and Scott [2]. None of these works treats the straight-sided Gjerde–Scott method, for which $h_\Gamma^{3/2}$ is the sharp target. New here are: (i) the leak law, with constant 1 and two-sided bounds (Theorem 1); (ii) the proof that the mean-free traction attains $h_\Gamma^{3/2} + h_\Omega^k$ for general data (Theorem 2), and that 3/2 is sharp for every $k \geq 4$ (Remark 1); (iii) the penalty threshold $\mu \asymp h_\Omega / \min_e |e|$, sufficient, and necessary under a shape hypothesis by a certified witness (Proposition 3); (iv) computations with $k = 4$ confirming all of this, including the penalty study requested in the prize. The failure of the naive and L_0^2 forms (Section 3) is elementary; compare [14, (5.7)–(5.9)], [17, Remark 3.2]. We treat a disk in a square; general domains, Clough–Tocher meshes, Navier–Stokes, 3D and omitted proofs are in [6]. The authors' respective uses of AI-assisted tools are declared at the end.

2. Setting and the three methods

Let $R = (-L, L)^2$, $L > 1$, D the unit disk, $\Omega = R \setminus \overline{D}$, $\Gamma = \partial D$, and let (u, p) be smooth (an assumption at the corners of R [6, Rem. 2.5]) with $-\Delta u + \nabla p = f$, $\operatorname{div} u = 0$ in Ω, $u|_\Gamma = 0$, $u|_{\partial R} = g$. Let P_h be a convex polygon with vertices on Γ and $0 \in \operatorname{int} P_h$, and assume h_Γ is sufficiently small. Set $\Gamma_h = \partial P_h$ and $\Omega_h = R \setminus P_h \supset \Omega$. On Γ_h and on Γ, n is the unit normal pointing out of Ω_h (resp. Ω); ν is the outward normal of R; $\langle \cdot, \cdot \rangle$ is the $L^2(\Gamma_h)$ product. A triangulation of Ω_h with edges $\mathcal{E}_\Gamma$ on Γ_h has $h = h_\Omega := \max \operatorname{diam} T$, $h_\Gamma := \max_{\mathcal{E}_\Gamma} |e|$, $\rho := h/\min_{\mathcal{E}_\Gamma} |e|$ and $\gamma := \mu h_\Gamma / h$, so $\mu/h = \gamma/h_\Gamma$. We assume (M0) shape regularity and $|e| \geq c_0 h_\Gamma$ on $\mathcal{E}_\Gamma$, (M1) no singular vertices, and (M2) the angle condition of [2, Thm. 1], with its angular parameter bounded below independently of the mesh size; $h_\Gamma \ll h$ is allowed. For $k \geq 4$, V_h is the space of continuous piecewise-P_k fields, V_h^R and V_h^0 are its subspaces vanishing on ∂R and on $\partial\Omega_h$, Π_h is the space of discontinuous piecewise-P_{k-1} functions, and $Z_h := \{v \in V_h^R : \operatorname{div} v = 0\}$. Then $\operatorname{div} V_h^R = \Pi_h$, and $(V_h^0, \Pi_h \cap L_0^2)$ has an inf-sup constant β independent of h [2], [6, Lemmas 4.3, 4.5]. With $Du = \nabla u + (\nabla u)^\mathsf{T}$, $\partial_n u = (\nabla u) n$ and $\bar{p}_{\Gamma_h}(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q$, the Nitsche form

$$N_h(u, v) = \tfrac{1}{2}(Du, Dv) - \langle \partial_n u, v \rangle - \langle u, \partial_n v \rangle + (\mu/h)\langle u, v \rangle$$

is the symmetric Nitsche form of [5]. All methods seek $u_h \in V_h$ with the outer data on ∂R and $p_h \in \Pi_h$ with $(q, \operatorname{div} u_h) = 0$ for $q \in \Pi_h$ and

$$N_h(u_h, v) - (p_h, \operatorname{div} v) + b_\Gamma(p_h, v) = (\tilde{f}, v) \qquad \forall v \in V_h^R. \tag{1}$$

The printed method GS [5, (28)] has $b_\Gamma = 0$ (we read its test space as V_h^R and use the opposite pressure sign; see [6, §2]). The naive form has $b_\Gamma = \langle p_h, v \cdot n \rangle$, the consistent traction term. The mean-free form CNS* has $b_\Gamma = \langle p_h - \bar{p}_{\Gamma_h}(p_h), v \cdot n \rangle$. The L_0^2 variant is the naive form with $p_h, q \in \Pi_h \cap L_0^2(\Omega_h)$. The outer data are the P_k interpolant g_I of g, or $\tilde{g}_I := g_I - (m_R/8L^2)x$ with $m_R := \int_{\partial R} g_I \cdot \nu$, which has zero net flux, like the exact data (cf. [19]). The velocities of GS and CNS* are divergence-free pointwise.

To compare u_h with u on $\Omega_h \supset \Omega$ we extend u as a divergence-free field. Let $\tilde{u} := \operatorname{curl} \tilde{\psi}$ for a smooth extension $\tilde{\psi}$ of the stream function of u, let $\tilde{p}$ and $\tilde{f}$ extend p and f, and let $F := \tilde{f} + \Delta\tilde{u} - \nabla\tilde{p}$, which vanishes on Ω. Set

$|\!|\!| v |\!|\!|^2 := \|\nabla v\|^2 + \frac{\mu}{h}\|v\|^2_{\Gamma_h} + \sum_{e\in\mathcal{E}_\Gamma} |e|\|\partial_n v\|^2_{L^2(e)}$ and $G := \|p - \bar{p}\|_{L^2(\Gamma)}$. Let $\varepsilon_h := h^k + \inf |\!|\!| \tilde{u} - z |\!|\!|$, the infimum over divergence-free $z \in V_h$ with the outer data. For bounded γ, $\varepsilon_h \le Ch^k$ with data $\tilde{g}_I$, and also with g_I if $h_\Gamma \ge h^4$ (k even) or $h_\Gamma \ge h^2$ (k odd) [6, Lemmas 4.6, 7.32]. C, c do not depend on $h, h_\Gamma, \mu, \gamma, \rho$; C_p also on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$.

The penalty must dominate the inverse trace constant on the shortest wall edge. With κ the Korn constant on V_h^R and C_I the inverse trace constant, if $\mu \ge 4\kappa C_I^2 \rho$ (true if $\gamma \ge \gamma_0 := 4\kappa C_I^2/c_0$), then $N_h(v,v) \ge c_1 |\!|\!| v |\!|\!|^2$ and $|N_h(w,v)| \le M |\!|\!| w |\!|\!| v |\!|\!|$ on V_h^R, with c_1, M independent of h, μ, ρ [6, Lemma 4.2].

3. Main results

The polygon enters through a consistency error $\mathcal{G}$, small because chords lie $O(h_\Gamma^2)$ from Γ and the normal tilt is odd on each edge.

Lemma 1 (Consistency)**.** *For $v \in V_h^R$, $N_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v\cdot n\rangle = (\tilde{f}, v) + \mathcal{G}(v)$, where $|\mathcal{G}(v)| \le C(1+\gamma^{1/2})h_\Gamma^{3/2} |\!|\!| v |\!|\!|$ and $\mathcal{G}(v) := \langle (\nabla\tilde{u})^{\mathsf{T}} n, v\rangle - \langle \tilde{u}, \partial_n v\rangle + \frac{\mu}{h}\langle \tilde{u}, v\rangle - (F, v)$.*

Proof. Use Green's formula. Chords lie within $h_\Gamma^2/4$ of Γ, so $\tilde{u} = O(h_\Gamma^2)$ on Γ_h and $|(F,v)| \le Ch_\Gamma^2 \|\nabla v\|$; this bounds the last three terms. On Γ, $(\nabla u)^{\mathsf{T}} n = n \operatorname{div} u = 0$. On a chord e the normal tilts by an angle $s + O(h_\Gamma^2)$, where s is the arclength from the midpoint m_e, so $(\nabla\tilde{u})^{\mathsf{T}} n = s\, a_e + O(h_\Gamma^2)$ with a_e constant on e. The remainder contributes $Ch_\Gamma^2 \|v\|_{L^1(\Gamma_h)}$. The leading term is $O(h_\Gamma)$ pointwise but odd in s: $|\int_e s\, a_e \cdot v| = |\int_e s\, a_e \cdot (v - v(m_e))| \le |e|^2 \|\partial_\tau v\|_{L^1(e)}$, and an inverse trace inequality gives $Ch_\Gamma^{3/2}\|\nabla v\|$. □

For $v \in Z_h$, $\int_{\Gamma_h} v\cdot n = \int_{\Omega_h} \operatorname{div} v = 0$, so for GS

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \qquad \mathcal{R}(v) := -\langle \tilde{p} - \bar{p}, v\cdot n\rangle \qquad \forall v \in Z_h. \tag{2}$$

So GS is driven by the missing load $\mathcal{R}$, and its response is, to leading order, the leak field $(h/\mu)v_\varphi$ defined next. Let $\varphi := \chi(r)\int_0^\theta (p - \bar{p})\, d\theta'$ in polar coordinates, with a cut-off χ (1 near $r = 1$, 0 near ∂R); then $w := \operatorname{curl}\varphi = -(p - \bar{p})n$ on Γ. With $I_h^{C^1}$ the C^1 piecewise-P_{k+1} interpolant [20], $v_\varphi := \operatorname{curl} I_h^{C^1}\varphi \in Z_h$, $\|v_\varphi - w\|_{W^{1,\infty}} \le Ch^k$, and $v_\varphi = -(p-\bar{p})(x^*)n + O(h_\Gamma + h_\Omega^k)$ on Γ_h, with x^* the radial projection of x onto Γ.

Theorem 1 (The printed method [6, §6])**.** *Let $\gamma \ge \gamma_0$ and $Y := (1+\gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$. (a) $|\!|\!| \tilde{u} - u_h |\!|\!| \le C[(h/\mu)^{1/2}G + Y]$. If $G > 0$ and $h \le h_0(G)$, with h_0 independent of γ, then $|\!|\!| \tilde{u} - u_h |\!|\!| \ge (2M)^{-1}(h/\mu)^{1/2}G - CY$. (b) $\|\nabla(\tilde{u} - u_h)\| \le C_p h/\mu + CY$. (c) Let $G > 0$, fix γ, and let $h \to 0$ with ρ bounded and $\varepsilon_h \le Ch^{3/2}$. Then $\tilde{u} - u_h = (h/\mu)v_\varphi + \vartheta_h$ with $|\!|\!| \vartheta_h |\!|\!| = o((h/\mu)^{1/2})$ and $\|\vartheta_h\|_{\Gamma_h} = o(h/\mu)$; hence*

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \to 1, \qquad \frac{|\!|\!| \tilde{u} - u_h |\!|\!|}{(h/\mu)^{1/2}G} \to 1, \qquad c\frac{h}{\mu}G \le \|\nabla(\tilde{u} - u_h)\| \le C_p \frac{h}{\mu}.$$

The H^1 rate in (b) is 1, not 1/2, because incompressibility turns the normal load into a tangential derivative, integrated by parts along the wall; an inverse estimate gives only $|\langle S, \partial_n v\rangle| \le Ch_\Gamma^{-1/2}\|\nabla v\|$.

Lemma 2 (Normal-load cancellation)**.** *Let τ_e, n_e be the unit tangent and normal of $e \in \mathcal{E}_\Gamma$, and S edgewise $W^{1,\infty}$ on Γ_h, continuous at vertices, with $|S\cdot\tau_e| \le Ch_\Gamma$ on each e. Then $|\langle S, \partial_n v\rangle| \le C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2}\|\nabla v\|) \le C\|\nabla v\|$ for $v \in Z_h$.*

Proof. The tangential part is at most $Ch_\Gamma \|\partial_n v\|_{\Gamma_h} \le Ch_\Gamma^{1/2}\|\nabla v\|$. On e, $\operatorname{div} v = 0$ gives $\partial_n v \cdot n_e = -\partial_\tau(v\cdot\tau_e)$, so with $\sigma_e := S\cdot n_e$, $\int_e \sigma_e\, \partial_n v\cdot n_e = \int_e \partial_\tau\sigma_e\,(v\cdot\tau_e) - [\sigma_e\, v\cdot\tau_e]_{\partial e}$. The integral is at most $C\|v\|_{L^1(e)}$. At a vertex $x = e\cap e'$ the endpoint terms combine to $v(x)\cdot(\sigma_{e'}\tau_{e'} - \sigma_e\tau_e) = O(|e| + |e'|)|v(x)|$, as S is continuous and the edge directions differ by $O(|e| + |e'|)$. A one-dimensional inverse inequality gives $\sum_e |e|\|v\|^2_{L^\infty(e)} \le C|\Gamma_h|^{1/2}\|v\|^2_{\Gamma_h}$, and $\|v\|_{\Gamma_h} \le C\|\nabla v\|$ uniformly in h [6, Lemma 3.2]. □

Proof of Theorem 1. As the projection $\Gamma_h \to \Gamma$ has Jacobian $1 + O(h_\Gamma^2)$, $|\mathcal{R}(v)| \le (G + Ch_\Gamma^2)(h/\mu)^{1/2} |\!|\!| v |\!|\!|$. (a) For the upper bound test (2) with $z - u_h \in Z_h$, z attaining ε_h. For the lower, $\mathcal{R}(v_\varphi) = G^2 + O(h_\Gamma^2 + h^k)$, $|\!|\!| v_\varphi |\!|\!| \le (\mu/h)^{1/2}(G + Ch_\Gamma^2 + Ch^k) + C$,

Table 1: H^1 seminorm errors (rate), tests A and B_{100}, and pressure errors $\|p^* - p_h\|_{L^2(\Omega_h)}$ for B_{100}. Strong: $u_h = 0$ at the nodes of Γ_h, mesh (a) (wall vertices in two or four triangles, violating (M2)). GS, CNS*: standard mesh. Rates are computed from the preceding mesh in $N \in \{16, 24, 32, 48, 64, 96, 128, 192\}$ using $r = \log(E_{\text{previous}}/E_{\text{current}})/\log(h_{\Omega,\text{previous}}/h_{\Omega,\text{current}})$; at $N = 256$, GS and CNS* give 1.55, 1.56 on test A.

	test A, velocity			test B_{100}, velocity		test B_{100}, pressure	
N	strong (a)	GS	CNS*	GS	CNS*	GS	CNS*
64	3.75e-1 (0.62)	1.81e-2 (1.68)	2.08e-2 (1.69)	1.88 (0.97)	6.83e-3 (1.76)	11.8 (0.65)	1.62e-2 (2.08)
96	2.98e-1 (0.59)	9.57e-3 (1.63)	1.09e-2 (1.64)	1.28 (0.98)	3.54e-3 (1.68)	9.23 (0.62)	8.01e-3 (1.80)
128	2.54e-1 (0.56)	6.11e-3 (1.59)	6.96e-3 (1.60)	9.71e-1 (0.98)	2.24e-3 (1.63)	–	–
192	–	3.27e-3 (1.57)	3.71e-3 (1.58)	6.54e-1 (0.99)	1.18e-3 (1.60)	–	–

and $M|\tilde{u}-u_h| \geq (\mathcal{R}(v_\varphi)-|\mathcal{G}(v_\varphi)|)/|v_\varphi|$. (b) Write $z-u_h = \delta_R+\delta_G$ in Z_h with $N_h(\delta_R,\cdot) = \mathcal{R}$ and $N_h(\delta_G,\cdot) = \mathcal{G}-N_h(\tilde{u}-z,\cdot)$ on Z_h, so that $|\delta_G| \leq CY$. For $r_h := \delta_R - (h/\mu)v_\varphi$ and $v \in Z_h$,

$$N_h(r_h, v) = -\langle(\tilde{p}-\bar{p})n + v_\varphi, v\rangle - \tfrac{h}{\mu}\big[\tfrac{1}{2}(Dv_\varphi, Dv) - \langle\partial_n v_\varphi, v\rangle - \langle v_\varphi, \partial_n v\rangle\big].$$

On e, $(\tilde{p}-\bar{p})n+v_\varphi = (p-\bar{p})(m_e^*)s\tau_e+O(h_\Gamma^2+h^k)$, with m_e^* the projection of m_e. This is odd in s to leading order, so the first term is at most $C_p(h_\Gamma^{3/2}+h^k)\|\nabla v\|$, as in Lemma 1. Lemma 2 with $S = w$ gives $|\langle v_\varphi, \partial_n v\rangle| \leq (C_p+Ch^k h_\Gamma^{-1/2})\|\nabla v\|$. Testing with $v = r_h$ gives $|r_h| \leq C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h$, and (b) follows. (c) Here $\vartheta_h = (\tilde{u}-z) + \delta_G + r_h$, so $|\vartheta_h| \leq C_p h/\mu + CY$, $\|\vartheta_h\|_{\Gamma_h} \leq (h/\mu)^{1/2}|\vartheta_h|$, and $\|v_\varphi\|_{\Gamma_h} = G + O(h_\Gamma^2 + h^k)$. Testing (2) with v_φ, all terms but $(\mu/h)\langle\tilde{u}-u_h, v_\varphi\rangle$ and $\mathcal{R}(v_\varphi)$ are $o(G^2)$ by (a), (b), so $\|\tilde{u}-u_h\|_{\Gamma_h} \geq c(h/\mu)G$; and $\|\tilde{u}-u_h\|_{\Gamma_h} \leq C\|\nabla(\tilde{u}-u_h)\| + Ch^{k+1/2}$ by a uniform trace inequality. □

Naive and L_0^2 forms. Adding the traction back naively fails. The naive system is square, and $(0, 1)$ lies in the kernel of its matrix, since $-(1, \operatorname{div} v) + \langle 1, v\cdot n\rangle = 0$ for $v \in V_h^R$. In the L_0^2 variant, $\operatorname{div} u_h \in \Pi_h$ is orthogonal to $\Pi_h \cap L_0^2$, so $\operatorname{div} u_h \equiv c$ with $c|\Omega_h| = \int_{\partial R} g_I \cdot \nu + \int_{\Gamma_h} u_h \cdot n$, and nothing forces $c = 0$. The mean-free traction avoids both defects: it ignores constants, so the constant pressure mode is fixed by $(p_h, \operatorname{div} v)$ as in GS, and the velocity stays divergence-free.

Theorem 2 (The mean-free method; [6, Thm. 7.2, Rem. 7.3])**.** *There is $\gamma_2 \asymp \max(\gamma_0, \beta^{-2})$ such that for $\gamma \geq \gamma_2$: (a)* CNS* *is uniquely solvable, and its velocity equals that of the naive form tested only with $\{v \in V_h^R : \int_{\Gamma_h} v\cdot n = 0\}$, as in [14]; (b) $|\tilde{u}-u_h| \leq C[(1+\gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$; (c) consequently, for $\gamma \leq \gamma_{\max}$ and data $\tilde{g}_I$, $\|\tilde{u}-u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$, with C depending on $\gamma_{\max}$ (also with g_I, see Section 2).*

Sketch. Let $B^*(q, v) := -(q, \operatorname{div} v) + \langle q - \bar{p}_{\Gamma_h}(q), v\cdot n\rangle$. Then $B^*(q+c, v) = B^*(q, v) - c\int_{\Gamma_h} v\cdot n$, and Lemma 1 gives $N_h(e_u, v) + B^*(e_p, v) = \mathcal{G}(v)$ on V_h^R, where $e_u := \tilde{u} - u_h$, $e_p := p^* - p_h$ and $p^* := \tilde{p} - \bar{p}_{\Gamma_h}(\tilde{p})$. Write $e_p = \eta + \xi$ with $\eta := p^* - \pi_h p^*$ (π_h the L^2 projection onto Π_h), and let $\bar{\xi}$ be the mean of ξ. Testing with V_h^0, the inf-sup condition gives $\beta\|\xi-\bar{\xi}\| \leq C(|e_u| + h_\Gamma^{3/2})$. Testing with $z - u_h \in Z_h$ leaves $\langle\eta+\xi-\bar{\xi}, (z-u_h)\cdot n\rangle$, and $|\langle\xi-\bar{\xi}, v\cdot n\rangle| \leq C(c_0\gamma)^{-1/2}\|\xi-\bar{\xi}\||v|$ by a discrete trace inequality; this is absorbed for $\gamma \geq \gamma_2$, which proves (b). For (a), zero data give $u_h = 0$, $p_h \equiv c$, and a flux-carrying edge bubble gives $c = 0$; on zero-flux tests B^* is the naive form. (c): use a Friedrichs inequality. □

Remark 1 (Further results that complete the picture, proved in [6])**.** (1) *Pressures, uniformity* [6, Thms. 8.11, 8.7, 7.30, 7.33]. The GS pressure error is $\asymp h_\Gamma^{1/2}G/\gamma$ up to $\pm C_p h_\Gamma$ (fixed γ, bounded ρ); with data $\tilde{g}_I$ that of CNS* is $O(h_\Gamma^{3/2} + h_\Omega^k)$, and Theorem 2(c) is uniform in $\gamma \geq \gamma_2$. (2) *Sharpness, every $k \geq 4$* [6, Thms. 7.8, 7.24]. Assume (S) $h_\Gamma \leq 2\sin(\theta_{\min}/4)$, $\theta_{\min}$ the smallest angle of the triangles touching Γ_h, and let h_Γ be small. If $W = \partial_n u \cdot \tau \not\equiv 0$ on Γ, then $\|\nabla(\tilde{u}-u_h)\| \geq c\|W\|h_\Gamma^{3/2} - Ch_\Gamma^2$ for CNS*, and for GS with any G, with c independent of γ, ρ, p. The idea, detailed in [6, Thm. 7.24]: for $v \in Z_h$ vanishing on Γ_h with $\int_e \partial_n v = 0$ edgewise, the pressure and penalty terms of N_h drop out, so coercivity is tested against $\mathcal{G}$ alone; the chordal bump $-s^2W_e/2$ in the consistency error of Lemma 1 then pairs with the second moment of $\partial_n v\cdot\tau_e$, and this pairing is made strictly positive, under (S), by an explicit C^1 piecewise-P_5 stream function supported on the three triangles meeting each wall vertex. (3) *A growing penalty* [6, Thm. 7.30, Cor. 7.34]. For GS with flux-corrected outer data $\tilde{g}_I$, a penalty $\mu \gtrsim h_\Omega h_\Gamma^{-3/2}$ yields $\|\tilde{u}-u_h\|_{H^1(\Omega_h)} \leq C_p(h_\Gamma^{3/2} + h_\Omega^k)$.

Scott asked how to choose the penalty. On one well-shaped triangle at a short wall edge, a divergence-free P_4 field has negative Nitsche energy unless $\mu|e|/h$ is large.

Table 2: Coercivity threshold μ^* and $\gamma^* = \mu^* h_\Gamma/h$, with h/h_Γ and ρ varied through the outer box $(-L, L)^2$ and Γ_h fixed. Under refinement at $L = 2.5$, $\gamma^* = 14.7, 18.1, 19.9, 20.8$ for $N = 8, 16, 32, 64$.

$N = 16$	L	2.5	5	10	20	40	$N = 32$	2.5	5	10	20	40
	ρ	4.70	8.47	16.2	31.8	63.0		5.73	9.98	18.7	36.3	71.5
	μ^*	59.3	112	208	411	809		66.0	114	218	422	836
	γ^*	18.1	19.0	18.4	18.5	18.4		19.9	19.7	20.1	20.0	20.1

Proposition 3 (Penalty threshold [6, §10])**.** *(a) $\mu \geq 4\kappa C_I^2 \rho$ suffices (Section 2). (b) Let T have an edge $e \subset \Gamma_h$ and not meet ∂R. Map it by a similarity to the triangle with base $[(0,0),(1,0)]$ and apex (a,b), with barycentric coordinates λ_0 (apex) and λ_1, λ_2. For $t \in \mathbb{R}^3$ let $v_t := \operatorname{curl}(\lambda_1^2\lambda_2^2(t_0\lambda_0 + t_1\lambda_1 + t_2\lambda_2)) \in P_4^2$. Let A_T, B_T, C_T be the quadratic forms $\int_T \frac{1}{2}|Dv_t|^2$, $\int_e \partial_n v_t \cdot v_t$ and $\int_e |v_t|^2$, and let λ_T be the top eigenvalue of the pencil $(2B_T - A_T, C_T)$. If $\mu|e|/h < \lambda_T$, then N_h is indefinite on Z_h for every $k \geq 4$. (c) $\lambda_T > 1$ if the apex angle is $\geq 35°$ and the base angles $\geq 5°$; $\lambda_T > 4$, > 9.3 for apex angles $\geq 45°$, $\geq 60°$. So if a triangle on a shortest edge of Γ_h has such a shape, coercivity on Z_h needs $\mu > \rho$.*

Proof. (b) The potential vanishes to second order on the edges through the apex, so v_t, extended by zero, lies in Z_h. A_T and B_T are invariant under dilation and C_T scales with $|e|$, so $N_h(v_t, v_t) = (\mu|e|/h - \lambda_T)C_T(t) < 0$ for the top eigenvector. (c) A_T, B_T, C_T are explicit rational functions of (a,b) [6, §10.1]. On dyadic boxes covering each shape region we verify $2520\, b^3 q^{\mathsf{T}}(2B_T - A_T - \lambda C_T)q > 0$ for a rational q in exact interval arithmetic [6, Prop. 10.3]. For tall triangles see [6, Prop. 10.6, Thm. 10.13]. One-triangle failure already occurs at a minimum angle of 11.3°, by an exact LDL^{T} certificate [6, §10.1]. In absolute, mesh-independent terms: if every triangle meeting $B(x_0, 3Y)$, for x_0 a wall-edge midpoint and $Y \leq 10^{-6}$, has diameter at most $\kappa_0(\theta)Y$ for the shape angle θ of that star (with $\kappa_0(30°) = 0.0117$), then N_h is indefinite on Z_h for $\mu < 15h/Y$. □

4. Numerical study

We take $k = 4$, $L = 2.5$, and N polygon vertices radially projected from equispaced points on the square; then $h/h_\Gamma \approx 3.3$, $\max|e|/\min|e| \leq 1.97$, and each vertex of Γ_h lies in three triangles. Direct solves give residuals below 10^{-12} and $\|\operatorname{div} u_h\| \leq 2 \cdot 10^{-9}$ (10^{-6} for strong, mesh (a)). We use $\mu = 100$ ($\gamma \approx 30$), known to be above the measured thresholds only (CNS*: by 27% at $N = 16$, 13% at $N = 64$, ≈7% extrapolated [6, Rem. 15.18]). Test A is Scott's benchmark $u = (1 - r^{-2})(-y, x)$, $p = 0$. Test B_a has stream function $(r^2 - 1)^2(x + y^2/2)/10$ and $p = a(x^2 y - y + x/2)$, so $G = |a|(7\pi/8)^{1/2}$.

In Table 1 strong imposition locks (rate → 1/2), but not on our standard mesh, which satisfies (M2) (rate 1.61 at $N = 128$); on test A ($G = 0$) both Nitsche methods tend to 3/2. On B_{100} the printed method has H^1 rate 1, an error over 500 times that of CNS* at $N = 192$, and pressure rate decreasing towards 1/2. On the pure-pressure test ($u = 0$, $f = \nabla p$, p as in B_1), CNS* gives $u_h = 0$ to round-off, and at $N = 96$ the ratios in Theorem 1(c) are 0.981, 0.996 ($\mu = 10^2$), 0.9993, 0.9994 ($\mu = 10^4$).

To test Proposition 3 we find μ^* by bisection on the inertia of the velocity block plus a divergence penalty. With Γ_h fixed and the box enlarged ($L = 2.5, \ldots, 40$), μ^* is linear in ρ over more than a decade ($\mu^* = 59.3, \ldots, 809$ for $\rho = 4.70, \ldots, 63.0$ at $N = 16$; similarly at $N = 32$), so $\gamma^* \approx 18$–21 and $\mu^*/\rho \approx 11$–13; under refinement at $L = 2.5$, $\gamma^* = 14.7, 18.1, 19.9, 20.8$ for $N = 8, \ldots, 64$. In the reported experiments, $\mu \geq 25\rho$ exceeded all measured thresholds, while a fixed μ fell below the measured threshold once $h_\Omega/h_\Gamma \gtrsim \mu/20$. Table 2 gives the full sweep.

CRediT authorship contribution statement

Jaideep Sai Padhi: Formal analysis, Software, Validation, Writing – original draft. **Claudemir de Souza Cavalcante:** Formal analysis, Writing – review & editing.

Declaration of competing interest

The question studied is the subject of a prize offered by L. R. Scott; the authors intend to submit this work for it. The authors declare no other competing interests.

Data availability

Code and archived numerical outputs supporting this study are available at https://github.com/jaideepsaipadhi/zero-gradient-prize.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work J. S. Padhi used Claude (Anthropic), a large language model, in order to develop and check proofs, write and run the numerical code, search and check the literature, and draft and edit the text. The proofs were checked by separate instances of the model acting as independent referees, the computer-assisted steps by exact-arithmetic certificates, and the numbers by reproducible public code (see Data availability). C. de S. Cavalcante used ChatGPT 6 Astra (OpenAI) to help locate relevant mathematical theorems, correct grammar, and write computer code. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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